Department of Computer Science & Engineering

Course Instructor: **Md. Asif Khan Rifat** Lab: **07**
Course: **CSE103 (Structured Programming)**

Topics Today
1. Strings

Strings in C are arrays of characters ending with a null terminator ('\0'). The <string.h> library provides several built-in functions to simplify common string operations.

Key String Functions in <string.h>

1. **strlen(s)**
   Returns the length of the string s (excluding the null terminator).
   Example: `strlen("hello") → 5`
2. **strlwr(s) / strupr(s)**
   Converts all characters in the string s to lowercase/uppercase.
   Example: `strlwr("Hello") → "hello"`
3. **strrev(s)**
   Reverses the string s.
   Example: `strrev("hello") → "olleh"`
4. **strcpy(dest, src)**
   Copies the content of src into dest.
   Example:

```
char src[] = "hello", dest[10];
strcpy(dest, src);  // dest now contains "hello"
```

5. **strncpy(dest, src, n)**
   Copies up to n characters from src into dest.
6. **strcat(dest, src)**
   Concatenates (appends) src to the end of dest.
   Example:

```
char str1[20] = "hello ", str2[] = "world";
strcat(str1, str2);  // str1 now contains "hello world"
```

7. **strcmp(s1, s2)**
   Compares two strings lexicographically.
   - Returns 0 if they are equal.
   - Returns a positive/negative value if s1 > s2 or s1 < s2.

Problem 1: Palindrome Checker
Write a program that takes a string as input and checks if it is a palindrome. A palindrome reads the same backward as forward. Ignore case and spaces.

Example:
Input: "A man a plan a canal Panama"
Output: "Palindrome"

Hint:

- Use strlwr() to convert the string to lowercase.
- Remove spaces by copying only non-space characters into a new string. Pseudocode is given below:

```
1.      char str[100], result[100];
2.      int i, j = 0;
3.      printf("Enter a string: ");
4.      gets(str);
5.      for (i = 0; i < strlen(str); i++) {
6.          if (str[i] != ' ') {
7.              result[j] = str[i];
8.              j++;
9.          }
10.           }
11.           result[j] = '\0';
12.           //now result contains string without spaces
```

- Reverse the string using strrev() and compare it with the original.

Problem 2: Anagram Detector
Write a program to check if two strings are anagrams of each other. Ignore case and spaces.

Example:
Input: "Listen" and "Silent"
Output: "Anagrams"

Hint:

- Use strlwr() to convert both strings to lowercase.
- Remove spaces from both strings. See Problem 1 hint to remove spaces.
- Sort both strings alphabetically. Sorting of a string pseudocode is as follows:

```
1. for(i=0; i<strlen(str)-1; i++){
2.          for(j=i+1; j<strlen(str); j++){
3.              if(str[i]>str[j]){
4.                  temp = str[i];
5.                  str[i] = str[j];
6.                  str[j] = temp;
7.              }
8.          }
9.      }
```

- Compare each character of both string at same index after sorting.

Problem 3: Word, letter, vowel, consonant Counter
Write a program to count the total words, letters, vowels and consonants in a given string. Ignore case and punctuation marks.

Example:
Input: "Hello world! Hello everyone."
Output:

Word: 4
Vowel: 9
Consonant: 14
Letter: 23

Hint:

- Use strlwr() to convert the string to lowercase.
- Remove punctuation/non-alphabetic characters using a loop. Pseudocode is below:

```
1.      for (i = 0; str[i] != '\0'; i++) {
2.          // Check if the character is not a non-alphabetic letter
3.          if ((str[i] >= 'A' && str[i] <= 'Z') || (str[i] >= 'a' &&
   str[i] <= 'z') || (str[i] == ' ') ||  (str[i] >= '0' && str[i] <=
   '9')){
4.              result[j] = str[i];
5.              j++;
6.          }
7.      }
```

- Count the words, letters, vowels and consonants. See lecture slide.

Problem 4: Substring Search
Write a program that searches for a substring in a string and prints its starting index.

Example:
Input:

- String: "the quick brown fox jumps over the lazy dog"
- Target: "brown"
  Output: "brown starts at index 10"

Hint:

- See lecture slide